Hierarchical Bilinear Scattering Transformer for Remote-Sensing Image Scene Classification

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***Abstract*—In remote-sensing scene classification (RSSC), persistent challenges such as high interclass similarity and substantial intraclass diversity remain key obstacles to accurate recognition. Although vision Transformer (ViT) has demonstrated outstanding performance, it tends to smooth out high-frequency discriminative details due to the inherent low-frequency preference of self-attention. To alleviate the feature oversmoothing induced by self-attention, we propose a novel hierarchical bilinear scattering Transformer (HBS-Former) that jointly captures local frequency-domain details and global semantic context. First, to alleviate the basis-dependent representation limitations of a single wavelet transform, a dual-stream scattering network is proposed by integrating the discrete wavelet transform (DWT) and the dual-tree complex wavelet transform (DTCWT). This design enables stable, direction-aware feature extraction in the frequency domain. Second, a global–local dual-path representation module is elaborately designed to capture long-range contextual information and fine-grained local details via global attention and local convolution branches, which effectively strengthens the complementarity of multigranularity feature representations. Furthermore, a hierarchical bilinear fusion (HBF) mechanism is developed to explicitly model second-order interactions among hierarchical features. By modeling pairwise bilinear interactions across hierarchical features, it produces more discriminative image representations. Extensive experiments on three public benchmarks, namely Aerial Image Dataset (AID), NWPU, and UCM, demonstrate that HBS-Former achieves a favorable tradeoff between classification accuracy and model complexity compared with existing RSSC methods. Code will be available at https://github.com/yizhilanmaodhh/HBS-Former**

***Index Terms*—Bilinear fusion, discrete wavelet transform (DWT), dual-stream frequency decomposition, dual-tree complex wavelet transform (DTCWT), global–local dual-path representation, remote-sensing scene classification (RSSC).**

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I. Introduction

REMOTE-SENSING imagery comprises observations of the Earth's surface and its surrounding environment acquired via RS technologies. With the rapid advances in Earth observation systems, the volume and diversity of available RS data have grown explosively [1]. A central challenge is, therefore, how to process, analyze, and interpret these data to extract meaningful insights about the Earth's surface, for which modern computer vision models provide powerful solutions. In this context, remote-sensing scene classification (RSSC) has emerged as a fundamental task in RS image understanding. By assigning semantic scene labels to aerial images, RSSC enables identification of land-surface scene types and provides critical evidence for characterizing the macroscopic spatial organization of complex regions. Owing to its scene-level semantic expressiveness, RSSC has been widely applied in environmental monitoring [2], agricultural production assessment [3], natural-resource management [4], and urban planning.

Early RSSC approaches predominantly relied on hand-crafted descriptors, such as scale-invariant feature transform (SIFT) [5], texture descriptors (TDs) [6], color histograms (CHs) [7], histograms of oriented gradients (HOGs) [8], and GIST [9]. These descriptors can extract useful cues from images. Among them, CH, GIST, and TD capture global characteristics and can be directly applied to scene classification. However, local descriptors such as SIFT and HOG are insufficient to represent an entire scene, which motivates the use of encoding schemes to aggregate local features into global representations. Representative methods include the improved Fisher kernel (IFK) [10], vectors of locally aggregated descriptors (VLADs) [11], spatial pyramid matching (SPM) [12], and bag-of-visual-words (BoVWs) [13]. Owing to their simplicity and efficiency, these pipelines are widely adopted and achieve substantial progress, particularly for scenes with clear category differences and relatively simple structures. However, the growth in image resolution has led to more complex scene categories, making fine-grained discrimination particularly challenging. In such settings, the representational capacity of hand-crafted features is no longer sufficient to support high-accuracy classification, calling for more advanced learning-based solutions.

The rapid development of deep learning, particularly the success of convolutional neural networks (CNNs), has

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propelled RSSC into a new stage [14], [15], [16]. Through hierarchical feature learning, CNNs can automatically discover rich and discriminative representations and have achieved remarkable success across computer vision tasks such as scene classification, semantic segmentation, and object detection. To enhance discriminability in complex scenes, extensive architectural innovations have been explored. For example, Zhang et al. [17] combined CNNs with capsule networks (CapsNet) to explicitly encode spatial orientation and hierarchical relations of key entities, improving recognition of categories that are visually homogeneous yet differ in layout. Sun et al. [18] proposed a global–local dual-branch encoding network (GLNet) that separately extracts whole image semantics and local region details. In parallel, feature representation and fusion strategies have been actively developed [19]. Yuan et al. [20] introduced pyramid multisubset weighted deep feature fusion to integrate multilevel CNN features. Yu et al. [21] adopted bilinear CNNs for RSSC, which extract abstract features with a lightweight MobileNetV2 backbone and form bilinear features [22] from transformed deep representations. Collectively, these studies have advanced CNN-based RSSC from fine-tuning a single backbone to more refined paradigms. Nevertheless, the fixed sampling patterns of conventional convolution may limit its adaptability to the complex geometric variations and spatial distributions of RS scenes [23], [24]. Therefore, more flexible modeling of spatial relationships and global context remains necessary.

More recently, inspired by the success of Transformers in natural language processing, vision Transformers (ViTs) have been rapidly deployed in computer vision. Dosovitskiy et al. [25] demonstrated that Transformers can model global dependencies and long-range relations effectively in image recognition. Subsequent work has extended ViT variants to RSSC [26]. Bazi et al. [27] provided an early application of ViTs to RS classification with encouraging results. Wang et al. [28] studied training Transformer backbones from scratch on large-scale RS datasets (e.g., MillionAID [29]) followed by downstream fine-tuning. However, Wang et al. [30] reported that self-attention exhibits an inherent low-pass filtering limitation. With increasing network depth, the stacking of self-attention layers progressively smooths high-frequency cues, such as edges and texture details. This smoothing effect reduces interclass separability and amplifies intraclass variability, thereby impairing discriminative performance in RS imagery. These observations have spurred two main research directions. One line pursues hybrid CNN–Transformer designs to harness their complementary strengths. CNNs excel at capturing local receptive field patterns and high-frequency details (e.g., textures and edges), whereas Transformers are effective at global context modeling and long-range semantic dependencies [31]. For instance, Zhang et al. [32] incorporated multihead self-attention (MHSA) bottlenecks into a ResNet backbone to balance local and global modeling. Guo et al. [33] proposed a channel–spatial attention Transformer (CSAT) that enhances feature representation through joint channel and spatial attention. The other line adopts a frequency-domain perspective, which introduces Fourier or wavelet transforms [34], [35], [36], [37] to better capture and exploit high-frequency information. Yang et al. [38] proposed SF-MSFormer, which fuses multiscale features across spatial and frequency domains, and designs an interpretable texture enhancement encoder. More recently, Yang et al. [39] further introduced cross-domain attention to enable bidirectional adaptive fusion between spatial representations and frequency-domain details.

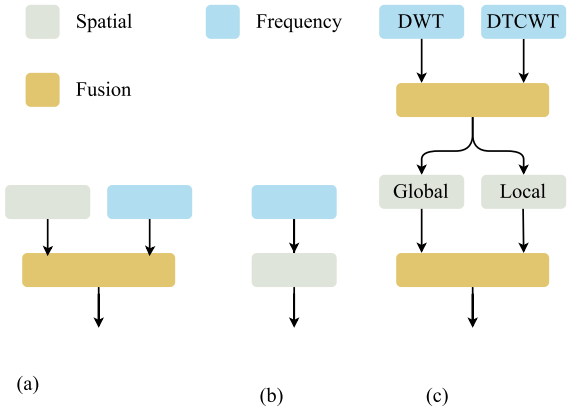


Fig. 1. Conceptual comparison of representative spatial–frequency Transformer paradigms. (a) Parallel spatial–frequency fusion represented by SF-MSFormer [38] and WSC-Net [39]. (b) Single-wavelet frequency modeling represented by SVT [40]. (c) Proposed dual-wavelet frequency modeling followed by global–local spatial modeling.

Despite these advances, two key limitations persist. First, existing methods mainly exploit the complementarity between spatial- and frequency-domain information, while usually relying on a single wavelet transform. Since different wavelet transforms possess distinct directional and shift-related properties, a single-transform design may not provide sufficiently comprehensive representations for diverse geometric structures in RS scenes. Second, most methods typically adopt shallow fusion paradigms to model multiscale feature interactions, which struggle to capture higher order cross-scale dependencies and latent feature correlations, thereby limiting their ability to improve interclass separability and intraclass compactness under complex scene variations.

To address the above gaps, we propose a hierarchical bilinear scattering Transformer (HBS-Former). First, rather than relying on a single wavelet transform [40], we introduce a novel dual-stream scatter block (DSS-Block). It combines the horizontal, vertical, and diagonal detail responses of the discrete wavelet transform (DWT) with the enhanced directional selectivity of the dual-tree complex wavelet transform (DTCWT). This synergy effectively preserves discriminative details while mitigating the smoothing of local high-frequency information. To further strengthen global–local synergy, we design a dual-branch feature extraction module, which includes parallel self-attention and convolution pathways to jointly model semantic context and fine-grained texture structures within large and small receptive fields, respectively. Thus, the complementarity between global semantics and local details is further enhanced. The conceptual distinction between HBS-Former and representative spatial–frequency Transformer paradigms is illustrated in Fig. 1. Furthermore, to integrate hierarchical information ranging from superficial edges to

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deep semantics, we propose a hierarchical bilinear fusion (HBF) mechanism. Rather than naively stacking hierarchical features, the HBF computes pairwise bilinear interactions between aligned features from different stages to model cross-stage spatial affinities between shallow structural cues and deep semantic representations. By moving beyond first-order aggregation to second-order multiplicative interactions, this design enables the discovery of synergistic feature interplays, substantially amplifying the model's capacity to distinguish highly similar scene categories.

The main contributions of this study are as follows.

1) We propose the HBS-Former framework, which provides an integrated and effective solution for RSSC. By coupling the feature characteristics of different layers, the DSS-Block, dual-branch feature extraction module, and HBF mechanism are progressively combined, which effectively improves intraclass adhesiveness and interclass separability, and achieves consistent accuracy gains for RSSC.
2) A novel DSS-Block is proposed. Unlike single-transform designs, this block combines the horizontal, vertical, and diagonal detail responses of DWT with the enhanced directional selectivity of DTCWT. This combination helps preserve discriminative high-frequency details and capture complementary structural information.
3) We develop a global–local dual-path feature aggregation (GLDPFA), which employs parallel attention and convolution pathways with complementary receptive fields, thereby enhancing local high-frequency details alongside global semantics in the spatial domains.
4) An HBF mechanism is proposed to elevate feature aggregation from linear combination to second-order statistical affinity modeling. By modeling pairwise second-order interactions across stages, this design preserves the topological integrity of scene representations, significantly enhancing discriminability for fine-grained interclass variations.

The remainder of this article is organized as follows. Section II introduces the frequency-domain representations, while Section III presents the proposed HBS-Former. Section IV reports the experimental results and analyses, and Section V concludes this article.

II. Background: Frequency-Domain Representations

A. Frequency-Domain Analysis via Wavelet Transforms

Frequency-domain approaches provide a complementary viewpoint to spatial-domain representations. By decomposing a signal into frequency components at different scales and orientations, they can characterize high-frequency details more directly, such as texture and edges, while partially reducing the influence of illumination changes and local perturbations. Consequently, wavelet transforms, as a canonical tool for multiscale analysis, have been widely used in image recognition [41]. The standard 2-D DWT [42] decomposes an image into one low-frequency subband (LL) for global contours and three high-frequency subbands (LH, HL, HH) for horizontal, vertical, and diagonal details. Owing to its orthogonal wavelet basis, DWT provides concentrated responses to horizontal, vertical, and diagonal structural variations. However, it lacks shift invariance and possesses limited directional selectivity.

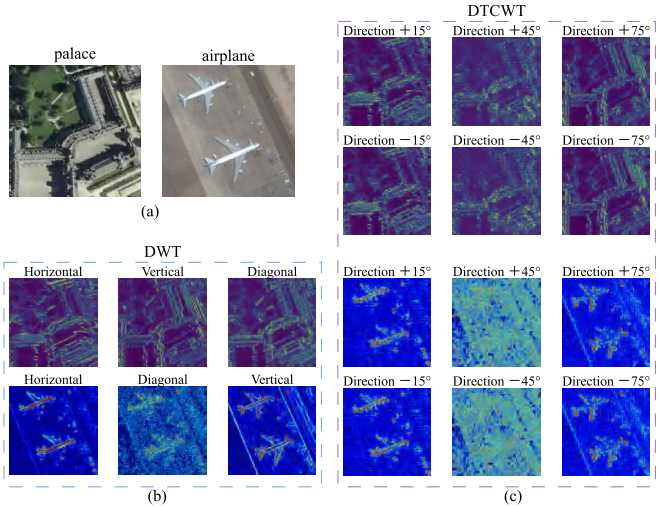


Fig. 2. Visual comparison of high-frequency subbands. (a) Original images. (b) DWT decomposition (orthogonal but shift-variant). (c) DTCWT decomposition (approximately shift-invariant and multidirectional).

To overcome these drawbacks, the DTCWT [43] employs two parallel real-valued DWT trees to produce the real and imaginary parts of complex wavelet coefficients. By designing the filter banks such that the corresponding wavelets form an approximate Hilbert-transform pair $\psi_h^b(t) \approx \mathrm{H}\{\psi_h^a(t)\}$, DTCWT yields an approximately analytic transform. This analytic nature endows DTCWT with approximate shift invariance, substantially alleviating sensitivity to spatial translations and ensuring highly stable feature representations. Furthermore, it generates six distinct high-frequency directional subbands ($\pm 15°, \pm 45°, \pm 75°$). This strong multidirectional selectivity is particularly advantageous for RS applications, as it effectively captures complex oriented structures such as farmland ridges, road networks, and building contours, thereby enhancing model discriminability. While DTCWT provides improved directional selectivity and approximate shift invariance, DWT can produce more concentrated responses to horizontal, vertical, and diagonal structural variations.

B. Motivation

RS scenes contain structures with diverse orientations, scales, and spatial arrangements. Since DWT and DTCWT possess different directional and shift-related properties, their representations may emphasize different structural cues across scene categories.

As visually compared in Fig. 2, DWT [see Fig. 2(b)] concentrates high-frequency energy into three orthogonal subbands. These subbands can produce concentrated responses to horizontal, vertical, and diagonal variations, which may benefit some scene categories containing repeated or axis-aligned structures. However, this coarse directional quantization is inherently shift-variant. For some categories containing structures beyond the principal DWT orientations, the DWT-only

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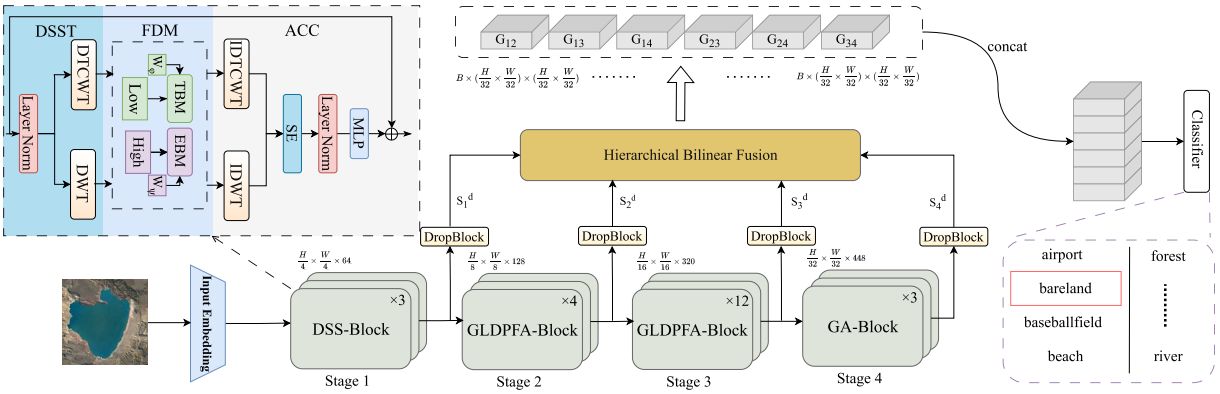


Fig. 3. Overall architecture of the proposed HBS-Former. The network consists of three key components: DSS-Block, GLDPFA-Block, and HBF, where GA-Block is standard global attention.

representation may be less effective, as reflected by its lower classwise performance for categories such as *palace* in our experiments. Conversely, DTCWT [see Fig. 2(c)] distributes geometric energy across six precise orientations ($\pm 15°, \pm 45°, \pm 75°$). While this dispersion might result in lower visual contrast for simple orthogonal edges compared to DWT, it mathematically preserves critical phase information and multidirectional stability. This provides orientation-selective representations that may complement DWT for some structures with varying or nonaxis-aligned orientations.

Our empirical findings, as detailed in Section IV, show that the DWT-only and DTCWT-only variants exhibit different category-dependent strengths. This empirical complementarity motivates our dual-stream architecture, which integrates the two frequency-domain representations to obtain a more comprehensive scene representation.

III. Methodology

In this section, we present the proposed HBS-Former in detail. It mainly includes four parts: DSS-Block, GLDPFA-Block, the GA-Block, and the HBF module. They are organically combined to achieve full coordination of local and global, as well as hierarchical features in the spatial and frequency domains. This enables the modeling of more complex interaction relationships between features and the capturing of more identifiable high-frequency details, effectively enhancing the model's expressive power and improving classification performance.

A. Overview

The overall architecture of HBS-Former is illustrated in Fig. 3. Patch Embedding is first used to divide an input image into nonoverlapping patches and is followed by linear projection to obtain initial feature representations. These features are progressively processed through a four-stage hierarchy.

DSS-Block (Stage 1): It performs dual-stream frequency-domain decomposition and reconstruction using DWT and DTCWT, thereby capturing complementary orthogonal and multidirectional structural representations and mitigating feature oversmoothing.

GLDPFA-Block (Stages 2 and 3): It models scene-level semantics via MHSA and local structures via depthwise separable convolution (DWconv), with outputs integrated by a learnable residual fusion mechanism. This block enhances the global–local representation in the spatial domain.

GA-Block (Stage 4): This block employs standard multihead attention to integrate and refine semantic features. This hierarchical arrangement naturally aligns with the feature transition in complex scenes. Early stages prioritize high-frequency spatial details, and intermediate stages couple local structures with broad contexts. Crucially, at deep low-resolution stages where discrimination relies predominantly on semantic aggregation, the architecture employs pure global attention (GA-Block) to maximize its direct benefits.

Finally, the output features from all four stages are aggregated into the HBF module, which explicitly captures pairwise second-order interactions across stages through bilinear interactions. The resulting discriminative hierarchical representation is then used for final classification.

B. Dual-Stream Scatter Block

To comprehensively capture complex geometric patterns and alleviate feature oversmoothing inherent in deep stacking, we propose the DSS-Block. As illustrated in Fig. 3, it processes the input spatial feature $X \in \mathbb{R}^{C\times H\times W}$ through three sequential phases, i.e., dual-stream scatter transformation (DSST), frequency-domain mixing (FDM), and adaptive channel calibration (ACC).

1) Dual-Stream Scatter Transformation: RS scenes contain structures with diverse orientations and spatial arrangements. To capture this structural diversity, X is simultaneously fed into two parallel scattering transformations using DWT and DTCWT, respectively, to capture complementary frequency-domain structural representations.

In the DWT scattering transformation, the standard DWT is applied to yield a low-frequency approximation subband $X_{ll}^{d} \in \mathbb{R}^{C\times H/2\times W/2}$ and three orthogonal high-frequency detail subbands $X_{h}^{d} \in \mathbb{R}^{C\times k\times H/2\times W/2}$ (horizontal, vertical, and diagonal). The horizontal, vertical, and diagonal DWT detail subbands respond to axis-aligned and local structural variations. In parallel, the DTCWT scattering

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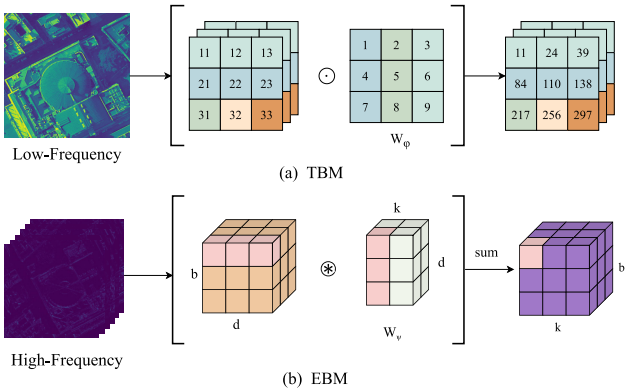


Fig. 4. Illustration of the proposed feature processing modules. (a) TBM modulates the low-frequency features via elementwise multiplication, while (b) EBM performs blockwise mixing of high-frequency features.

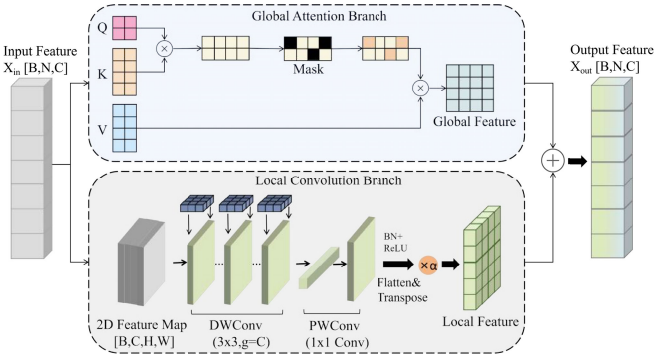


Fig. 5. Architecture of the GLDPFA module, consisting of parallel global-attention and local-convolution branches, fused via a learnable residual mechanism.

transformation performs an approximately analytic complex wavelet transform, generating a low-frequency component $X_{ll}^c \in \mathbb{R}^{C\times H/2\times W/2}$ and six orientation-specific high-frequency subbands $X_h^c \in \mathbb{R}^{C\times 2k\times H/2\times W/2\times 2}$ $(\pm 15°, \pm 45°, \pm 75°)$ along the real and imaginary components. This dual-stream decomposition separates low-frequency approximation information from high-frequency structural details. The formula can be simplified as

$$X_{ll}^d, X_h^d = \mathcal{F}_{DWT}(X) \tag{1}$$

$$X_{ll}^c, X_h^c = \mathcal{F}_{DTCWT}(X). \tag{2}$$

2) Frequency-Domain Mixing: To efficiently extract frequency-domain features, we use different methods to blend different frequency components by introducing learnable weights, as detailed in Fig. 4. For notational simplicity, X_{ll} denotes either X_{ll}^d or X_{ll}^c, while X_h denotes either X_h^d or X_h^c, depending on the corresponding stream.

For the low-frequency components X_{ll}^d and X_{ll}^c which encode global contours, the tensor blending method (TBM) is adopted to achieve effective global feature extraction. TBM uses elementwise tensor multiplication (known as the Hadamard product) to blend X_{ll} with a learnable spatial weighting matrix W_ϕ as

$$\tilde{X}_{ll} = [X_{ll} \odot W_\phi], \text{ where } W_\phi \in \mathbb{R}^{C\times H\times W}. \tag{3}$$

For the high-frequency components X_h^d and X_h^c, it is complex valued and have higher dimensions. To reduce the computational complexity while capturing fine-grained information, we employ the Einstein blending method (EBM) to blend X_h with learnable weight parameters W_ψ. The high-frequency components X_h is first reshaped from $\mathbb{R}^{m\times H/2\times W/2\times C}$ to $\mathbb{R}^{m\times H/2\times W/2\times C_b\times C_d}$, where $m = k$ for DWT, $m = 2 \times 2k$ for DTCWT, $C = C_b \times C_d$, and $b >> d$. The mixing along the last two dimensions is performed as follows:

$$\tilde{X}_h = [X_h \circledast W_\psi], \text{ where } W_\psi \in \mathbb{R}^{C_b\times C_d\times C_d}. \tag{4}$$

This operation explicitly enhances fine-grained geometric cues by capturing more directional features in the high-frequency domain.

3) Adaptive Channel Calibration: The mixed representations are mapped back to the spatial domain via the corresponding inverse scattering transform

$$\hat{X}_{DWT} = \mathcal{I}_{DWT}(\tilde{X}_{ll}^d, \tilde{X}_h^d) \tag{5}$$

$$\hat{X}_{DTCWT} = \mathcal{I}_{DTCWT}(\tilde{X}_{ll}^c, \tilde{X}_h^c). \tag{6}$$

The reconstructed features are concatenated as $X_{cat} = [\hat{X}_{DWT}, \hat{X}_{DTCWT}]$, which is then fed into ACC to calibrate the reconstructed dual-stream frequency features. In particular, ACC adopts the squeeze-and-excitation (SE) module [45] to generate input-adaptive channel weights from X_{cat}. This calibration enables the model to emphasize informative complementary responses from the DWT and DTCWT streams while suppressing redundant frequency channels.

C. GLDPFA Block

At high resolution, frequency-domain decoupling can effectively separate low-frequency structures from high-frequency details, while simultaneously characterizing holistic and local cues. However, as network depth increases and feature resolution decreases to a mid/low scale, global semantics and local details become highly coupled, making it unsuitable to continue modeling them independently. In such cases, excessive reliance on frequency-domain decomposition may instead weaken their synergy and limit cross-scale interactions. Prior studies [44] have also shown that, in complex RS scenes, frequency-domain features alone cannot fully preserve spatial structures, whereas spatial-domain features are often more advantageous for semantic modeling and detail characterization. Therefore, the GLDPFA-Block is designed. As illustrated in Fig. 5, this module contains two pathways: the global attention path models long-range semantic dependencies, while the local convolution path extracts fine details (e.g., textures and edges). The outputs of the two paths are integrated by a learnable residual fusion method to balance global and local information.

1) Global Branch: The global branch employs MHSA to capture long-range dependencies and aggregate contextual information in RS images. Given an input feature tensor

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$X_G \in \mathbb{R}^{B\times N\times C}$, the query, key, and value representations for the ith attention head are defined as

$$Q_i = X_G W_i^Q, \quad K_i = X_G W_i^K, \quad V_i = X_G W_i^V \tag{7}$$

where $W_i^Q \in \mathbb{R}^{C\times d_k}$, $W_i^K \in \mathbb{R}^{C\times d_k}$, and $W_i^V \in \mathbb{R}^{C\times d_v}$ are learnable projection matrices, and d_k and d_v denote the dimensions of the key and value representations, respectively. The output of the ith attention head is computed as

$$\text{Head}_i = \text{Softmax}\left(\frac{Q_i K_i^\top}{\sqrt{d_k}}\right) V_i. \tag{8}$$

The outputs of all attention heads are then concatenated and projected to obtain the MHSA output

$$Y_G = \text{Concat}\left(\text{Head}_1, \ldots, \text{Head}_h\right) W^O \tag{9}$$

where h denotes the number of attention heads and $W^O \in \mathbb{R}^{hd_v\times C}$ is the output projection matrix. By attending to the input features in multiple representation subspaces, the global branch captures contextual dependencies at different levels.

2) Local Branch: The local branch focuses on extracting local detail cues such as textures, edges, and small objects to provide fine-grained spatial representations. In our design, the local convolution pathway is implemented using DWconvs, which operate within a limited receptive field and are specialized for capturing local spatial relationships among neighboring pixels. For an input feature map $X_L \in \mathbb{R}^{B\times C\times H\times W}$, DW convolution followed by pointwise convolution (PWConv) is applied as

$$Y_L = \text{ReLU}(\text{BN}(\text{PWConv}(\text{DWConv}(X_L)))). \tag{10}$$

After convolution, the feature map $Y_L \in \mathbb{R}^{B\times C\times H\times W}$ is flattened back into a sequence form $\mathbb{R}^{B\times N\times C}$.

3) Learnable Residual Fusion: The learnable residual fusion is used to dynamically balance the contributions of local and global features. In particular, a learnable residual scaling factor a is utilized to regulate the influence of the local branch on the global representation. At initialization, a is small, and the model mainly relies on global semantics. As training proceeds, a gradually increases, allowing local detail cues to play a larger role and ensuring effective collaboration in the final representation. The fusion is defined as

$$Y_F = Y_G + a \cdot Y_L. \tag{11}$$

Overall, the collaborative modeling of these three parts ensures that the information at different scales is effectively captured and fused, thereby improving classification performance, especially for complex fine-grained scenes.

D. Hierarchical Bilinear Fusion

Pooling and prediction are typically performed by conventional classifiers directly based on high-level features from the final stage. However, scene separability in RS often depends simultaneously on both semantic layout and fine-grained details. In particular, fine-grained details are more conducive to distinguishing confusable classes. In shallow layers, these fine-grained details are more salient and tend to be suppressed as depth increases due to semantic aggregation. Therefore, the HBF mechanism is introduced before the classification head to jointly model deep semantic representations alongside multiscale features from shallow to deep layers, thereby further supplementing detail-related discriminative cues.

As illustrated in Fig. 3, the output feature map from the ith stage is denoted by $S_i \in \mathbb{R}^{H_i\times W_i\times C_i}$, where $i \in \{1, 2, 3, 4\}$. We first apply DropBlock [46] regularization to each stage feature, with a block size of $bs = 3$ and a drop probability of $p = 0.1$. Before bilinear interaction, the regularized feature S_i^d is spatially aligned to a common resolution $H' \times W'$ and projected to a common channel dimension d

$$\bar{S}_i = \mathcal{A}_i(S_i^d) \tag{12}$$

where $\mathcal{A}_i(\cdot)$ denotes a joint spatial and channel alignment operator that maps $S_i^d \in \mathbb{R}^{H_i\times W_i\times C_i}$ to $\bar{S}_i \in \mathbb{R}^{H'\times W'\times d}$.

Next, the resulting features are fed into a bilinear interaction unit (BIU) to perform the bilinear interaction calculation as shown in Fig. 6. Each BIU takes two feature maps from different stages, $\bar{S}_i$ and $\bar{S}_j$ $(i < j)$, as follows.

1) Spatial Flattening: The aligned feature map $\bar{S}_i$ is flattened along the spatial dimensions to obtain $\bar{S}_i^F \in \mathbb{R}^{H'W'\times d}$. In our experiment, $H'W' = 49$ and $d = 512$, and thus $\bar{S}_i^F \in \mathbb{R}^{49\times 512}$.

2) Bilinear Interaction: The matrix outer product is performed between the two flattened feature matrices to generate a bilinear interaction matrix $G_{ij} \in \mathbb{R}^{(H'W')\times(H'W')}$ as

$$G_{ij} = \bar{S}_i^F (\bar{S}_j^F)^\top. \tag{13}$$

Deviating from conventional first-order fusion strategies (e.g., elementwise addition or channel concatenation) that merely superimpose isolated responses, this mechanism goes beyond linear aggregation by explicitly computing pairwise cross-stage bilinear interactions between hierarchical representations. While first-order features only indicate the independent presence of basic patterns, the outer product formulation mathematically constructs pairwise multiplicative interactions across channels. Conceptually, this precisely quantifies the cross-stage spatial affinities of heterogeneous elements. In other words, it explicitly binds shallow high-frequency geometric edges with deep global semantic layouts at the same spatial coordinates. Consequently, this leap to second-order statistical affinities produces a highly geometry-aware representation.

Since $i, j \in \{1, 2, 3, 4\}$ and $i < j$, a total of $C_4^2 = 6$ distinct interaction matrices (e.g., $G_{12}, G_{13}, \ldots, G_{34}$) can be generated, each of which is computed by BIU. Finally, all six bilinear interaction matrices $\{G_{ij} \mid i < j\}$ are concatenated along the channel dimension and fused into a unified feature representation T via the integration function f as

$$T = f(G_{ij} \mid i < j). \tag{14}$$

The fused feature T integrates cross-stage information ranging from shallow textures to deep semantics, which is used for the final classification. By exploiting cross-stage bilinear affinities, the fused representation integrates shallow structural cues with deep semantic information, thereby improving discrimination among highly similar scene categories.

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TABLE I
STATISTICS OF THE RSSC DATASETS

Dataset	Classes	Total Images	Images/Class	Image Size	Spatial Resolution (m)
NWPU	45	31500	700	256×256	0.2–30
AID	30	10000	220–420	600×600	0.5–8
UCM	21	2100	100	256×256	0.3

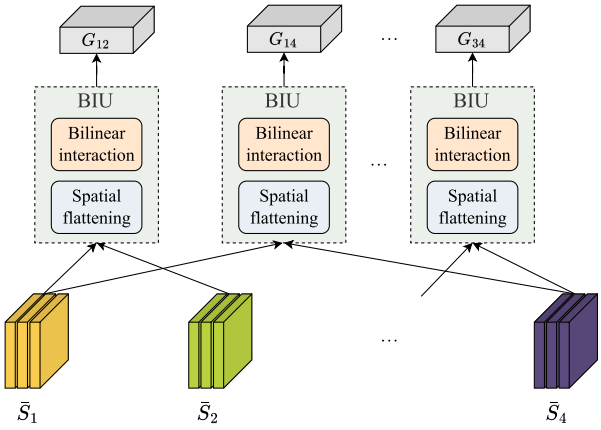


Fig. 6. Architecture of the proposed HBF module with pairwise bilinear interactions across hierarchical stage features.

IV. EXPERIMENT

A. Datasets

To comprehensively evaluate the effectiveness and generalization of the proposed model, we conduct extensive experiments on three representative public benchmark datasets for RSSC, namely NWPU-RESISC45 (NWPU), Aerial Image Dataset (AID), and UC Merced Land Use (UCM). Their scales and key statistics are summarized in Table I.

1) NWPU-RESISC45: NWPU [47] was released by Northwestern Polytechnical University and is one of the largest and most challenging datasets for RSSC. It contains 45 scene categories with 700 images per class, totaling 31 500 images. All images are of size 256×256 pixels, and the spatial resolution spans a wide range (0.2–30 m). The dataset covers complex and easily confusable scenes, such as airports, oil fields, overpasses, and stadiums. Example images from NWPU are shown in Fig. 7.

2) Aerial Image Dataset: AID [48] was jointly constructed by Wuhan University and Huazhong University of Science and Technology and includes 30 scene categories with approximately 220–420 images per class, totaling 10 000 images. Each image has a size of 600×600 pixels, and the spatial resolution ranges from 0.5 to 8 m. The dataset covers typical scenes such as airports, industrial areas, rivers, and commercial districts. Example images from AID are shown in Fig. 8.

3) UC Merced Land Use (UCM): The UCM [13] dataset was proposed by the University of California, Merced, and contains 21 semantic categories (e.g., agricultural areas, dense residential areas, and storage tanks), with 100 images per class and 2100 images in total. The image size is 256×256 pixels, with a spatial resolution of approximately 1 ft ($\approx$ 0.3 m). Example images from UCM are shown in Fig. 9.

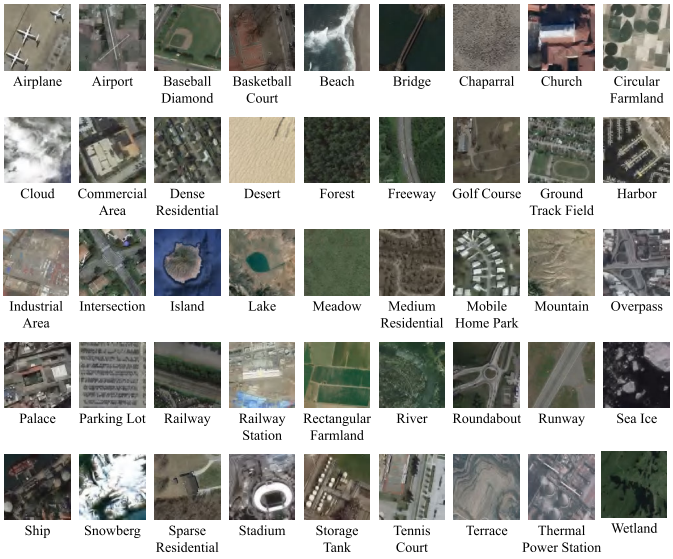


Fig. 7. Example images from the NWPU dataset.



Fig. 8. Example images from the AID dataset.



Fig. 9. Example images from the UCM dataset.

B. Implementation Details

Our model training follows a two-stage pipeline consisting of pretrained-parameter initialization and downstream fine-tuning. In the initialization stage, SVT [40] weights pretrained on ImageNet-1K are loaded to initialize the DTCWT branch and the GA block. Other components, including the DWT scattering layers and the HBF module, are randomly initialized. In this way, an initial model is obtained that combines strong visual priors with the proposed modules.

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TABLE II

COMPARISON OF MODEL COMPLEXITY AND PARAMETER COUNTS AMONG THE EVALUATED METHODS

Methods	Year	Params (M)	FLOPs (G)
ViT-B [25]	ICLR 2021	86.4	17.5
EMTCAL [53]	TGRS 2022	27	4.23
SKAL [54]	TNNLS 2022	-	-
CSAT [33]	Sci. Rep. 2022	86.0	16.9
CDLNet [55]	TGRS 2023	103.0	-
EMSCNet [56]	TGRS 2023	173.64	-
MSGNet [57]	TGRS 2023	10.35	-
MGML [58]	TNNLS 2023	-	-
RingMo-lite [59]	TGRS 2024	28.29	4.49
UPetu [60]	TGRS 2024	-	-
HFAM [61]	TGRS 2024	44.28	-
CGINet [62]	TGRS 2024	26.10	-
CGFNet [63]	TGRS 2024	-	-
Evo RSFormer [64]	TCSVT 2024	49.1	9.7
MSCN [65]	TGRS 2025	-	-
ACTFormer [66]	TGRS 2025	23.04	-
STMSF [67]	2025	48.85	-
HBS-Former	**2026**	**29.33**	6.68

TABLE III

CLASSIFICATION METRICS (OA%) OF HBS-FORMER AND COMPARED METHODS ON NWPU DATASET

Method	Training ratio	
	10%	20%
ViT-B [25]	91.59 ± 0.19	93.90 ± 0.20
EMTCAL [53]	91.63 ± 0.19	93.63 ± 0.12
SKAL [54]	90.04 ± 0.15	92.79 ± 0.11
CSAT [33]	89.70 ± 0.18	93.06 ± 0.16
CDLNet [55]	93.43 ± 0.18	95.30 ± 0.03
EMSCNet [56]	93.58 ± 0.07	95.37 ± 0.07
MSGNet [57]	92.40 ± 0.16	94.57 ± 0.12
MGML [58]	92.26 ± 0.08	94.27 ± 0.04
RingMo-lite [59]	89.85	93.25
UPetu [60]	92.20 ± 0.07	93.70 ± 0.14
HFAM [61]	90.81 ± 0.11	92.93 ± 0.17
CGINet [62]	92.28 ± 0.17	94.38 ± 0.13
CGFNet [63]	92.24 ± 0.22	94.42 ± 0.10
Evo RSFormer [64]	94.06 ± 0.05	95.72 ± 0.12
MSCN [65]	92.64 ± 0.09	94.59 ± 0.11
ACTFormer [66]	92.85 ± 0.14	94.76 ± 0.21
STMSF [67]	92.88 ± 0.16	94.95 ± 0.11
HBS-Former(Ours)	$\mathbf{97.32 \pm 0.18}$	$\mathbf{98.43 \pm 0.17}$

During fine-tuning, the initialized model is directly fine-tuned end-to-end for 300 epochs on each dataset. For data splits, NWPU adopts two train/test ratios of 1:9 and 2:8; AID uses 2:8 and 5:5; and UCM uses 5:5 and 8:2. Samples are randomly selected in a class-balanced manner, and the training and test sets are strictly separated without image overlap.

All experiments were performed on a workstation with Ubuntu 20.04 operating system, Intel i9-13900K CPU, 128 GB RAM, and NVIDIA RTX4090 graphics card. We use the AdamW optimizer [49] with an initial learning rate of 1×10^{-4}, a batch size of eight, and a cosine annealing schedule that adjusts the learning rate dynamically. The first five epochs are set for warm-up with the fixed learning rate 1×10^{-6}. To enhance generalization, RandAugment [50] and Mixup [51] are employed during training. Model performance is evaluated using overall accuracy (OA).

C. Comparison With State-of-the-Art Models

To verify the effectiveness and superiority of the proposed HBS-Former for RSSC, we compare it with seventeen existing methods on three benchmark datasets. The compared methods (summarized in Table II) cover major research directions in the field, including ViT-based models (ViT-B [25]), CNN–Transformer hybrids (EMTCAL [52], ACTFormer [65]), multiscale feature fusion approaches (MSCN [64], CGINet [61], CGFNet [62]), and attention-enhanced architectures (CSAT [33], HFAM [60]). All experiments follow the same data splitting and evaluation protocols. Each experiment is repeated five times with random seeds $\{0, 1, 2, 3, 4\}$, and the mean and standard deviation of OA are reported to ensure fairness and reproducibility.

Furthermore, an analysis of model complexity is conducted based on the parameter counts and FLOPs detailed in Table II. It is observed that conventional large-scale models, such as ViT-B and EMSCNet, entail massive parameter burdens exceeding 86M. On the contrary, the proposed HBS-Former maintains a small parameter scale and FLOPs of 29.33M and 6.68G, respectively. Coupled with the outstanding accuracy reported in subsequent evaluations, this indicates that a superior tradeoff between classification performance and model complexity is achieved by the proposed method.

1) Result on NWPU Dataset: NWPU is regarded as one of the most challenging benchmarks in RSSC. With 45 categories and pronounced interclass similarity, it imposes stringent requirements on a model's discriminative capability. As listed in Table III, HBS-Former demonstrates a substantial performance advantage under low training ratios of 10% and 20%. With 10% training data, HBS-Former achieves an OA of 97.32%, outperforming the classical ViT-B baseline (91.59%) by 5.73% and surpassing all contemporary or recent specialized models by a substantial margin. For example, it exceeds the efficient hybrid architecture EMTCAL (91.63%, + 5.69%), the strong multiscale fusion model MSCN (92.64%, + 4.68%), and the high-performing attention-synergy model ACTFormer (92.85%, + 4.47%). Our method also markedly outperforms CGINet (92.28%, + 5.04%) and CGFNet (92.24%, + 5.08%), which are specifically designed to enhance discriminability. When the training ratio is increased to 20%, HBS-Former further improves to 98.43% and remains consistently competitive. It outperforms the second-best method Evo RSFormer (95.72%) by 2.71%, and yields gains of 3.67% and 3.84% over ACTFormer (94.76%) and MSCN (94.59%), respectively.

Fig. 10(a) and (d) presents the confusion matrices of HBS-Former on NWPU under 10% and 20% training ratios, respectively. At 10% training, most misclassifications occur among a few category pairs with highly similar structures and semantic elements, such as the evident bidirectional confusion between palace and church, as well as confusion between railway and railway station. These errors typically arise because, under the nadir aerial view, the two classes share similar building/linear structures and texture patterns. In low data regimes, the models are more likely to rely on shared cues while overlooking subtle differences. When the training ratio increases to 20%, these confusions are substantially alleviated. Most notably, the recognition accuracy of the railway class

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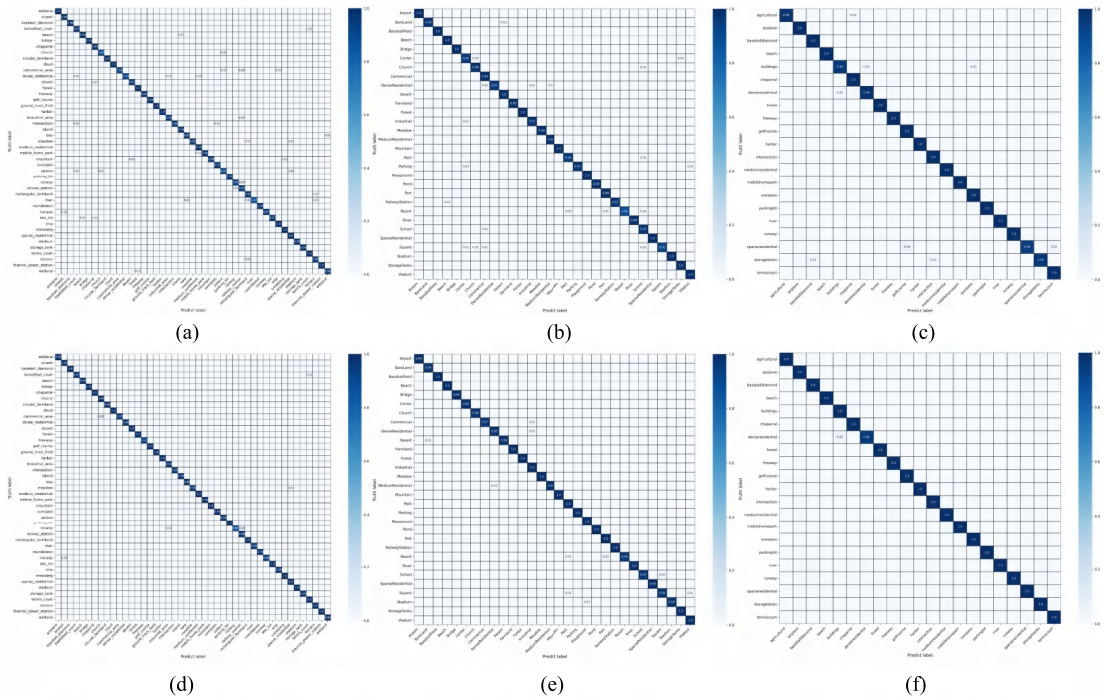


Fig. 10. Confusion matrices of HBS-Former on the NWPU, AID, and UCM datasets. (a) 10% training ratio on NWPU45. (b) 20% training ratio on AID. (c) 50% training ratio on UCM. (d) 20% training ratio on NWPU45. (e) 50% training ratio on AID. (c) 80% training ratio on UCM.

TABLE IV
CLASSIFICATION METRICS (OA%) OF HBS-FORMER AND COMPARED METHODS ON AID DATASET

Method	Training ratio	
	20%	50%
ViT-B [25]	94.90 ± 0.29	96.49 ± 0.18
EMTCAL [53]	94.69 ± 0.14	96.41 ± 0.23
SKAL [54]	94.38 ± 0.10	96.76 ± 0.20
CSAT [33]	92.55 ± 0.28	95.44 ± 0.17
CDLNet [55]	96.07 ± 0.16	97.31 ± 0.12
EMSCNet [56]	96.02 ± 0.18	97.35 ± 0.17
MSGNet [57]	95.46 ± 0.21	97.18 ± 0.16
MGML [58]	95.47 ± 0.06	97.05 ± 0.10
RingMo-lite [59]	93.86	96.54
UPetu [60]	95.33 ± 0.10	97.04 ± 0.21
HFAM [61]	94.71 ± 0.06	96.73 ± 0.04
CGINet [62]	95.34 ± 0.14	97.10 ± 0.24
CGFNet [63]	95.20 ± 0.14	97.10 ± 0.10
Evo RSFormer [64]	96.80 ± 0.14	98.24 ± 0.18
MSCN [65]	95.86 ± 0.16	97.46 ± 0.12
ACTFormer [66]	96.29 ± 0.19	97.56 ± 0.24
STMSF [67]	96.15 ± 0.16	97.51 ± 0.37
HBS-Former(Ours)	$\mathbf{97.96 \pm 0.22}$	$\mathbf{99.05 \pm 0.08}$

improves, and the confusion with railway station decreases markedly. Nevertheless, a small number of hard cases remain, such as occasional confusion between runway and airport in some samples.

2) Result on AID Dataset: The AID dataset provides a medium number of training samples while exhibiting a wide range of spatial resolutions, which jointly imposes comprehensive requirements on a model's scale adaptability and semantic understanding. As reported in Table IV, with a 20% training ratio, HBS-Former achieves an OA of 97.96%, substantially outperforming the classical ViT-B baseline (94.90%, + 3.06%). Compared with the recent high-performing hybrid model EMTCAL (94.69%, +3.27%) and the attention-enhanced model HFAM (94.71%, + 3.25%), HBS-Former shows consistent improvements. More importantly, it also surpasses the strong multiscale fusion model MSCN (95.86%, + 2.10%) and the attention-synergy model ACTFormer (96.29%, + 1.67%). When the training ratio is increased to 50%, HBS-Former further improves to 99.05% and continues to rank first overall. It exceeds the second-best Evo RSFormer (98.24%) by 0.81%, and outperforms the similarly competitive MSCN (97.46%) and ACTFormer (97.56%) by 1.59% and 1.49%, respectively. These results indicate that, as more training data become available, the HBS-Former architecture can more fully learn robust and discriminative joint representations, demonstrating strong scalability.

Fig. 10(b) and (e) shows the confusion matrices on AID dataset under 20% and 50% training ratios, respectively. With 20% training data, a few categories remain challenging. In particular, the resort class exhibits the lowest recall and is primarily misclassified as park, with additional minor confusions into School and RailwayStation. This can be attributed to the fact that resorts and parks often share visual elements in RS imagery, such as vegetation, water bodies, roads, and recreational facilities, which makes them more easily confused when their semantic compositions overlap. Meanwhile, square is occasionally predicted as urban functional categories such as center, church, and commercial, likely because squares and city centers often present large areas of hardened pavement and regular building boundaries, leading to limited local texture differences. When the training ratio increases to 50%, these confusions are markedly alleviated. In particular, the recognition rate of resort rises to approximately 0.96,

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TABLE V
CLASSIFICATION METRICS (OA%) OF HBS-FORMER AND COMPARED METHODS ON UCM DATASET

Method	Training ratio	
	50%	80%
ViT-B [25]	98.48 ± 0.39	99.01 ± 0.21
EMTCAL [53]	98.67 ± 0.16	99.57 ± 0.28
SKAL [54]	98.15 ± 0.05	99.52 ± 0.24
CSAT [33]	95.72 ± 0.23	97.86 ± 0.16
CDLNet [55]	99.06 ± 0.15	99.71 ± 0.09
EMSCNet [56]	99.10 ± 0.20	99.71 ± 0.17
MSGNet [57]	99.03 ± 0.11	99.76 ± 0.14
MGML [58]	99.07 ± 0.17	99.81 ± 0.18
RingMo-lite [59]	-	99.05
UPetu [60]	98.95 ± 0.12	99.05 ± 0.28
HFAM [61]	98.35 ± 0.14	99.43 ± 0.13
CGINet [62]	99.14 ± 0.16	99.84 ± 0.16
CGFNet [63]	-	99.76 ± 0.23
Evo RSFormer [64]	99.14 ± 0.19	99.76 ± 0.14
MSCN [65]	99.26 ± 0.07	$\mathbf{99.95 \pm 0.10}$
ACTFormer [66]	99.14 ± 0.09	99.76 ± 0.26
STMSF [67]	99.01 ± 0.31	99.58 ± 0.23
HBS-Former(Ours)	$\mathbf{99.37 \pm 0.09}$	99.86 ± 0.14

with only minor residual misclassifications as park (≈ 0.01) and RailwayStation (≈ 0.02). Concurrently, the recognition rate of square increases to approximately 0.98, with most other categories approaching saturation. Overall, with more training samples, the model can better learn fine-grained structural and contextual cues for distinguishing similar scenes (e.g., resort versus park and square versus city center), thereby effectively reducing misclassification among easily confused category pairs.

3) Result on UCM Dataset: UCM is a relatively small dataset and is, therefore, often used to evaluate a model's generalization ability and learning efficiency under limited training data. As listed in Table V, with a 50% training ratio, HBS-Former achieves an OA of 99.37%, delivering excellent performance. It is slightly superior to the current top-performing methods MSCN (99.26%) and CGINet (99.14%), and also outperforms ACTFormer (99.14%). When the training ratio increases to 80%, HBS-Former reaches an OA of 99.86%, remaining in the top tier. Although slightly below the best-performing MSCN (99.95%), it still clearly surpasses the general-purpose ViT-B baseline (99.01%) as well as several specialized models, such as EMTCAL (99.57%) and HFAM (99.43%). On the small-scale dataset, where accuracy tends to saturate, the OA gaps among advanced methods are typically very small. HBS-Former consistently stays within the best-performing group under both training ratios, indicating that it can stably learn effective scene representations from limited samples and maintain performance comparable to state-of-the-art approaches on highly saturated benchmarks.

Fig. 10(c) and (f) presents the confusion matrices on UCM dataset under 50% and 80% training ratios, respectively. With 50% training data, almost all categories are classified correctly, except for a few classes. The primary errors occur for dense residential, with approximately 5% of its samples misclassified as buildings. This is likely because both scenes exhibit dense building textures, similar roof patterns, and limited contextual cues. When the training ratio increases to 80%, overall misclassification is further reduced to only a few similar category pairs. For example, mutual confusion between dense residential and buildings persists. In addition, the recognition rate of sparse residential is relatively lower (≈ 0.94), with a small portion misclassified as golf course (≈ 0.04) and tennis court (≈ 0.02). Such errors often stem from local appearance similarities characterized by regular fields/grass textures, low-density buildings, or road boundaries. Overall, the remaining errors mainly arise from a very small number of category pairs with highly similar structures and textures, which is consistent with the typical behavior of datasets whose accuracy is close to saturation.

D. Ablation Analysis

1) Effectiveness of Key Components: As illustrated in Table VI, we define five network variants (Net1 to Net5) to systematically evaluate the contribution of key components in the proposed HBS-Former. First, to validate the effectiveness of the scattering module, we compare Net1 (using standard attention) and Net2 (utilizing DTCWT scatter). Replacing standard attention with the scattering method generally improves performance on most datasets; for instance, OA on the NWPU dataset (1:9) increases from 91.15% to 92.12%, and on the AID dataset (5:5) it increases from 96.24% to 98.26%.

The independent contribution of the GLDPFA-Block is evaluated through the Net3 variant. Table VI reveals that the effectiveness of this module varies with dataset scale and difficulty. In particular, the OA of Net3 on NWPU is 91.25% and 93.10% under the 1:9 and 2:8 splits, while on AID it is 95.01% and 96.30% under the 2:8 and 5:5 splits. These results are overall slightly lower than Net2. On the contrary, Net3 achieves 99.76% on UCM. This suggests that, for more challenging category sets, the GLDPFA-Block alone does not consistently yield improvements. However, on small-scale datasets where model performance tends to saturate, the advantage of local-structure modeling in the GLDPFA pathway is more likely to translate into noticeable gains.

Moreover, introducing the HBF module is the primary driver of further performance gains. Compared with Net2, adding HBF only at the high-level Stage 4 (Net4) yields limited improvements. On the NWPU dataset, the OA increases from 92.12%/94.50% to 92.18%/94.53% under the 1:9 and 2:8 splits, respectively. On the AID dataset (2:8), the OA improves from 95.49% to 96.00%. Similarly, the OA on the UCM dataset (8:2) rises from 98.33% to 99.29%. When HBF is extended to Stages 1–4 (Net5), the gains become more substantial. In particular, our method yields 96.14% and 96.31% OA on the NWPU dataset under the 1:9 and 2:8 splits. For the AID dataset, it achieves 97.58% and 98.76% OA (2:8 and 5:5), while attaining 99.52% OA on the UCM dataset (8:2). These results indicate that applying HBF across stages is more beneficial for integrating shallow details with deep semantics compared with introducing interactions only at high-level features. Consequently, feature discriminability is markedly strengthened.

Finally, the complete HBS-Former that incorporates the DSS-Block, GLDPFA-Block, and Stages 1–4 HBF achieves

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TABLE VI
ABLATION STUDY OF KEY COMPONENTS ON OA (%)

Variants	Methods			Datasets				
	Scatter	GLDPFA	HBF	NWPU (1:9)	NWPU (2:8)	AID (2:8)	AID (5:5)	UCM (8:2)
Net1				91.15	93.57	94.89	96.24	99.05
Net2	✓			92.12	94.50	95.49	98.26	98.33
Net3	✓	✓		91.25	93.10	95.01	96.30	99.76
Net4	✓		✓	92.18	94.53	96.00	98.12	99.29
Net5	✓		✓	96.14	96.31	97.58	98.76	99.52
HBS-Former (w/DWT)	✓	✓	✓	96.67	97.56	97.89	99.00	99.76
HBS-Former (w/DTCWT)	✓	✓	✓	96.57±0.14	97.48±0.19	97.94±0.12	98.95±0.18	99.81±0.19
HBS-Former(Ours)	✓	✓	✓	**97.32±0.18**	**98.43±0.17**	**97.96±0.22**	**99.05±0.08**	**99.86±0.14**

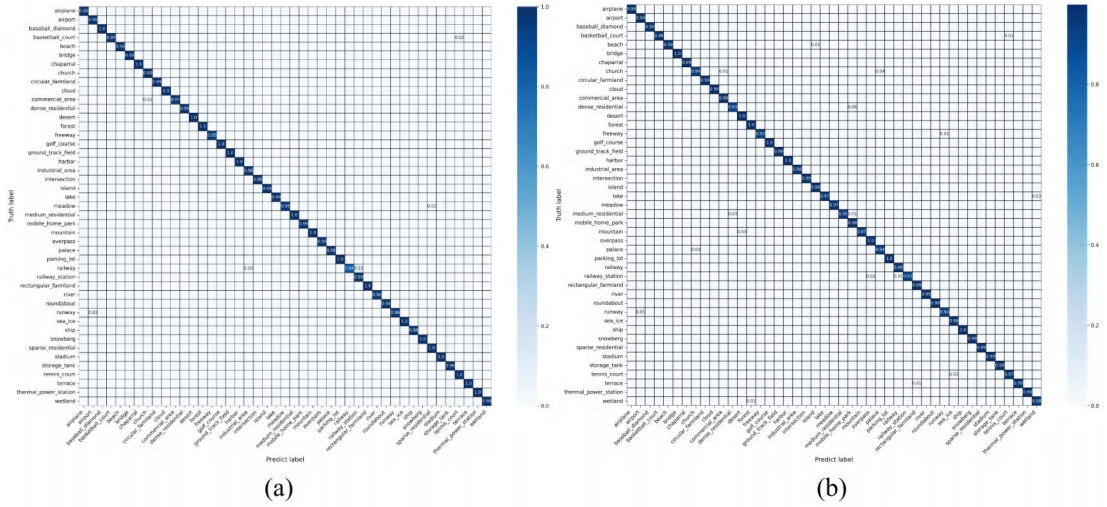


Fig. 11. Confusion matrices on the NWPU dataset (2:8 split) for (a) HBS-Former and (b) HBS-Former (w/DTCWT) variant.

the highest performance across all three datasets. In particular, it reaches 97.32% and 98.43% OA on the NWPU dataset under the 1:9 and 2:8 splits, while attaining 97.96% and 99.05% OA on AID under the 2:8 and 5:5 splits, and 99.86% OA on the UCM dataset. Compared with the Net5 variant using only cross-stage HBF, the complete HBS-Former achieves further improvements. For example, on the NWPU dataset (2:8), the OA increases from 96.31% to 98.43% (+2.12%), while on the AID dataset (5:5), the OA improves from 98.76% to 99.05% (+0.29%). These results demonstrate that the GLDPFA-Block and cross-stage HBF are not simply additive. Instead, these components are complementary, and their combination yields more stable and superior classification performance across varying dataset scales and split settings.

2) Effectiveness of Dual-Stream Scattering: To further validate the necessity of our dual-stream scattering transform, we compare two variants in Table VI, i.e., HBS-Former only with DWT (w/DWT), HBS-Former only with DTCWT (w/DTCWT). Although the DWT and DTCWT variants achieve competitive OA, their relative performance varies across different scene categories. The DWT-only variant performs better on several categories containing repeated or axis-aligned structures, while the DTCWT-only variant performs better on several categories containing directionally varying structures; however, this tendency is not universal across all categories. By jointly integrating the two frequency-domain representations, the dual-stream design alleviates the representational limitations of relying on either individual transform. As shown in Table VI, the dual-stream architecture achieves the highest OA in all five evaluated settings (e.g., 97.32% on NWPU 1:9 and 99.05% on AID 5:5). These results indicate that DWT and DTCWT provide complementary frequency-domain representations in most of the evaluated settings.

To visually and intuitively demonstrate the superiority of the dual-stream scattering transform, we compare the confusion matrices of the pure DTCWT variant [HBS-Former (w/DTCWT)] and the proposed dual-stream HBS-Former on the NWPU dataset (2:8 split), as shown in Fig. 11. The pure DTCWT variant [see Fig. 11(b)] struggles with scene categories dominated by orthogonal, grid-like layouts, such as dense_residential (0.93) and medium_residential (0.95), often confusing them with each other. Furthermore, it exhibits noticeable misclassifications between complex geometric structures like church (0.94) and palace (0.94). By introducing the DWT stream, the dual-stream HBS-Former [see Fig. 11(a)] incorporates additional horizontal, vertical, and diagonal detail responses. The accuracy of dense_residential and medium_residential surges to 0.99 and 1.0, respectively. Similarly, the confusion between church and palace is almost eliminated, with accuracies increasing to 0.99 and 0.98. These results further support the complementarity of the two frequency-domain representations for highly confusable scenes.

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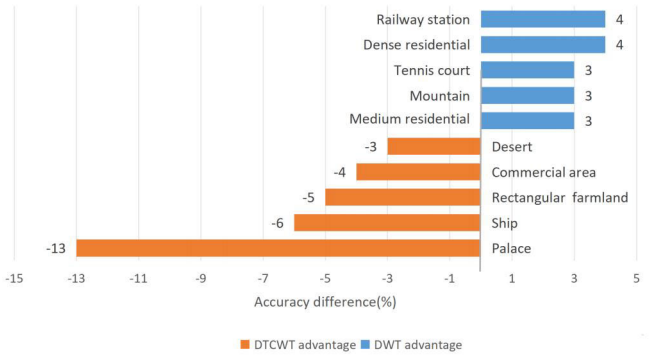


Fig. 12. Classwise accuracy differences between the DWT-only and DTCWT-only variants on NWPU under the 20% training ratio. The difference is calculated according to (15). Positive and negative values indicate relative advantages of DWT and DTCWT, respectively. For clarity, only categories with an absolute difference of at least three percentage points are displayed.

1) *Classwise Complementarity Analysis:* To quantify the empirical difference, let A_c^{DWT} and A_c^{DTCWT} denote the accuracies of the DWT-only and DTCWT-only variants for class c, respectively. We define the classwise accuracy difference as

$$\Delta_c = A_c^{\mathrm{DWT}} - A_c^{\mathrm{DTCWT}} \tag{15}$$

where a positive value indicates a relative advantage of the DWT-only variant, whereas a negative value indicates a relative advantage of the DTCWT-only variant.

As shown in Fig. 12, the two single-stream variants exhibit different classwise performance patterns. The DWT-only variant outperforms the DTCWT-only variant by four percentage points on railway station and dense residential, and by three percentage points on tennis court, mountain, and medium residential. On the contrary, the DTCWT-only variant achieves relative advantages of 13 percentage points on palace, six percentage points on ship, five percentage points on rectangular farmland, four percentage points on commercial area, and three percentage points on desert.

These results suggest that DWT provides relative advantages for several categories containing repeated layouts and prominent orthogonal boundaries, whereas DTCWT performs better for several categories containing directionally varying structures. Nevertheless, this relationship does not represent a strict one-to-one division between regular and irregular scenes. For example, DTCWT also achieves better performance on rectangular farmland. Therefore, the results are more appropriately interpreted as empirical evidence that the two transforms provide different and complementary representations across scene categories.

2) *Intermediate Feature Analysis:* To further examine the complementarity of the two streams at the feature level, Fig. 13 visualizes the intermediate responses extracted from the last DSS-Block of Stage 1. The dense residential and palace samples are selected as representative categories in which the DWT-only and DTCWT-only variants show relative advantages, respectively. To enable a fair visual comparison, the paired DWT and DTCWT branch responses are displayed using the same normalization range.

For the dense residential scene, the DWT stream produces relatively concentrated responses around repeated building rows, regular roofs, and road boundaries, whereas the DTCWT stream additionally responds to curved roads and locally varying directional structures. For the palace scene, the DWT stream mainly responds to several long and regular architectural boundaries, while the DTCWT stream exhibits broader responses around the courtyard and architectural structures with varying orientations.

In the relative-dominance maps, red and blue regions appear at different spatial locations, indicating that the two streams emphasize different spatial patterns within the same scene. The fused-response maps provide a qualitative visualization of the spatial responses retained after the two branches are integrated. These observations are consistent with the classwise quantitative results and provide additional interpretability for the dual-stream design.

TABLE VII
IMPACT OF THE GLDPFA-BLOCK STAGE CONFIGURATION ON MODEL PARAMETERS AND CLASSIFICATION ACCURACY

GLDPFA-Block	Params (M)	NWPU (2:8)	AID (5:5)	UCM (8:2)
Stages 2–3	**28.69**	**97.48**	**98.95**	**99.81**
Stages 2–4	29.31	97.36	98.72	99.76

E. Parameter Analysis

1) Effect of Stage Configurations in the GLDPFA-Block: We analyze the influence of deploying the GLDPFA-Block at different stages under the DTCWT-only configuration. As shown in Table VII, deploying the GLDPFA-Block in Stages 2–3 achieves better classification accuracy with fewer parameters. In particular, the Stage 2 and Stage 3 configurations obtain 97.48%, 98.95%, and 99.81% OA on NWPU, AID, and UCM, respectively, with 28.69M parameters. When the GLDPFA-Block is further extended to Stage 4, the number of parameters increases to 29.31M, while the OA decreases to 97.36%, 98.72%, and 99.76% on the three datasets, respectively. This indicates that introducing GLDPFA into the deeper low-resolution stage does not consistently improve performance. Since Stage 4 mainly contains high-level semantic information, the GA-Block is more suitable for semantic aggregation at this stage.

2) Effect of the HBF Channel Dimension d: The common projection dimension d used to align the channel dimensions of the multistage features in the HBF module controls the representation capacity of hierarchical bilinear interaction. To investigate its influence, we evaluate four settings, i.e., $d \in \{448, 512, 768, 1024\}$, on NWPU, AID, and UCM. As shown in Fig. 14, when $d = 512$, the model achieves the best performance on all three datasets, with OA values of 98.43%, 99.05%, and 99.86%, respectively. A smaller channel dimension may limit the representation capacity of cross-stage

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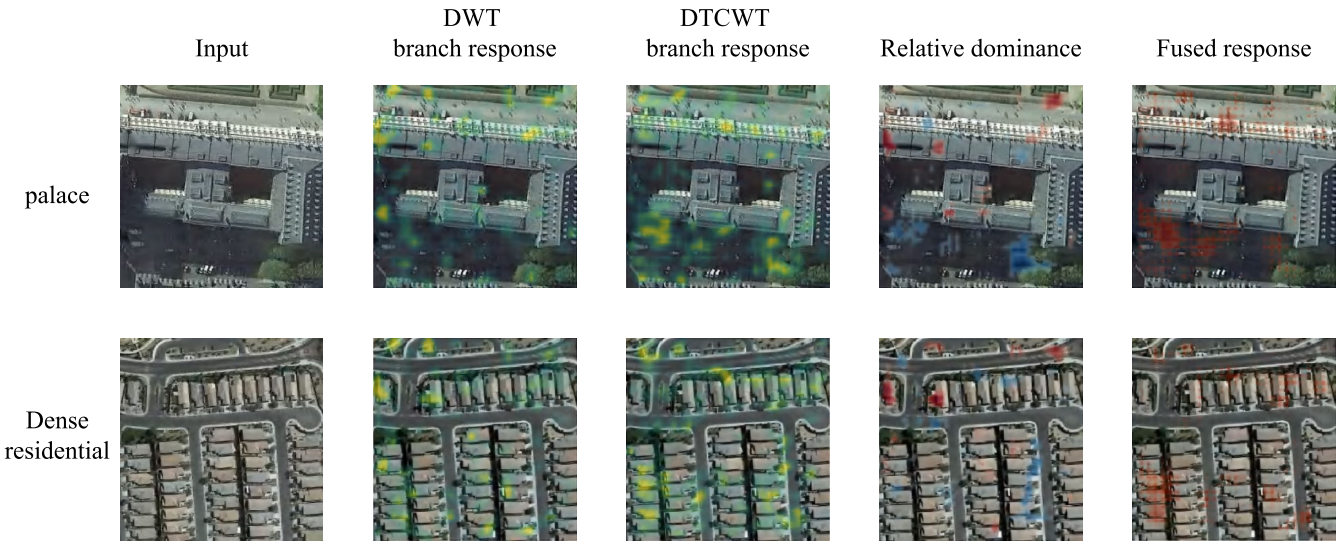


Fig. 13. Intermediate feature responses of the DWT and DTCWT streams within the DSS-Block for representative palace and dense-residential scenes.

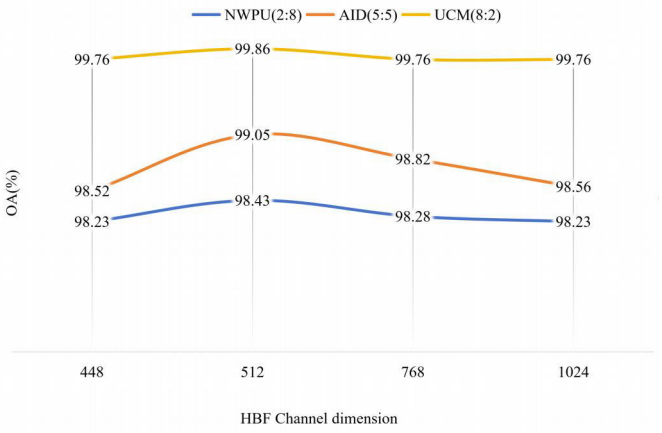


Fig. 14. Effect of the HBF channel dimension d on classification accuracy across three datasets.

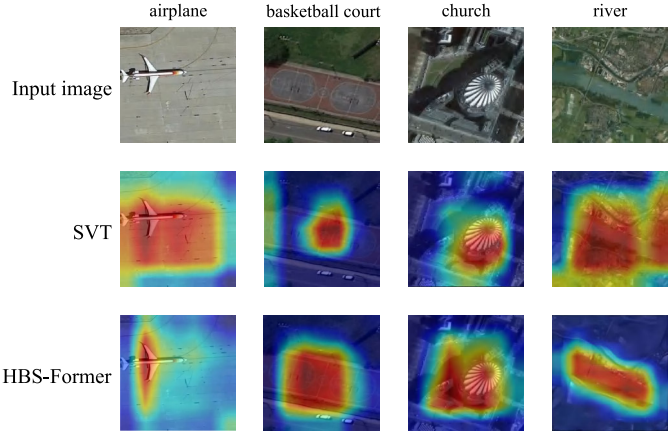


Fig. 15. Grad-CAM++ results on NWPU. HBS-Former achieves more precise target localization and background suppression than SVT.

second-order interaction, while a larger channel dimension does not bring further improvement and may introduce redundant correlations. For example, when d increases from 512 to 1024, the OA on AID decreases from 99.05% to 98.56%. Therefore, $d = 512$ is adopted in the final model to balance classification accuracy and model complexity.

F. Visualization

To more intuitively demonstrate the effectiveness and interpretability of the proposed HBS-Former, we utilize Grad-CAM++ [67] to generate visual explanations. This method highlights crucial regions for predicting the target, allowing for intuitive observation of the model's predictive focus. We selected four categories from the test set of the NWPU dataset, namely airplane, basketball_court, church, and river. As shown in Fig. 15, it can be observed that, compared to the baseline SVT, HBS-Former enables more accurate localization of key areas and effectively suppresses background clutter.

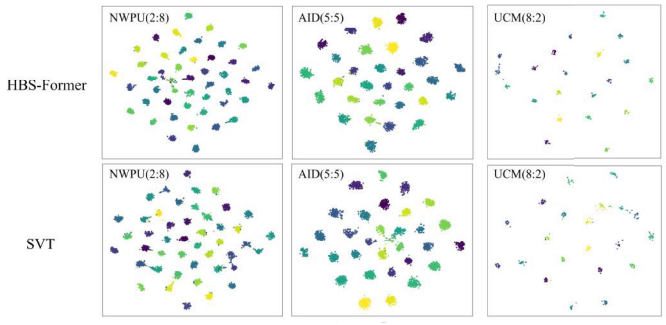


Fig. 16. t-SNE comparison on three datasets. (Top) HBS-Former yields tighter clusters and clearer class separation than (bottom) SVT.

This confirms the model's superior capability in extracting discriminative local features from complex RS scenes.

Beyond local saliency, we further evaluate the global distribution of the learned representations using t-SNE [68]. As shown in Fig. 16, a t-SNE visualization comparison is conducted between the test-set features of HBS-Former and

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SVT under the settings of the NWPU dataset (2:8), AID dataset (5:5), and UCM dataset (8:2). Compared with SVT, the feature distributions with a more pronounced class-clustering pattern are generated by HBS-Former. In particular, intraclass compactness is enhanced while interclass boundaries become more distinct and class overlap is minimized. This superiority is particularly evident on the NWPU dataset (2:8), where different categories are highly similar, indicating that more discriminative representations are learned by the proposed model. Under the higher training-ratio settings of the AID dataset (5:5) and UCM dataset (8:2), a consistent trend of compact intraclass clusters and separated interclass distributions is observed. These results align with the performance gains reported in the quantitative evaluations.

V. Conclusion

In this study, we present HBS-Former for RSSC. This framework effectively mitigates self-attention-induced high-frequency information loss, so that improves identification ability in complex scenes by exploiting complementary spatial–frequency and hierarchical representations. The dual-stream scattering design combines DWT and DTCWT to preserve diverse structural responses while alleviating the representational limitations of relying on a single wavelet transform. Meanwhile, global–local dual-path modeling coordinates fine-grained local structures with long-range semantic context, while HBF further exploits cross-stage complementary information through explicit second-order interactions. Extensive experiments on NWPU, AID, and UCM demonstrate that these complementary mechanisms work collaboratively and enable HBS-Former to achieve consistently competitive classification performance with a favorable balance between accuracy and model complexity. Nevertheless, the additional frequency-domain processing and cross-stage interactions introduce extra computational overhead, and the current evaluation is mainly limited to optical RSSC benchmarks. Future work will focus on extending the HBS-Former framework to other remote-sensing tasks, such as scene segmentation and change detection, to further validate its generalizability. In addition, we aim to explore more efficient variants of HBS-Former to reduce computational overhead while maintaining high discriminative power, making the model more suitable for real-time applications on resource-constrained platforms.

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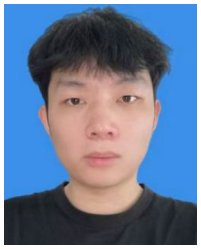
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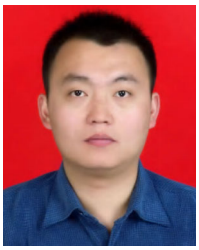
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